

Schema Renaming

Schema renaming lets you create a version of an existing schema with some component names changed. The notation $S[\text{new/old}]$ is per Z RM §3.11.

<i>Counter</i>
$count : \mathbb{N}$
$count \geq 0$

Single rename: produce a Counter with component "count" renamed to "n".

$$CounterN \triangleq Counter[n/count]$$

Multiple renames in one step:

<i>Point</i>
$x : \mathbb{N}$
$y : \mathbb{N}$

Rename both components at once.

$$SwappedPoint \triangleq Point[x/y, y/x]$$

Renaming Primed Schemas

The decoration is tightest-binding per Z RM §3.11: $S'[a/b]$ renames the primed schema S' , not the result of renaming then priming.

$$Op2 \triangleq Counter[count'/count]$$

In an operation schema context, this captures the after-state renaming: the existing component count is renamed to the primed component count' in the view.

Decorated Component Names

The rename pairs themselves may carry decoration.

$$CounterOp \triangleq Counter[count'/count]$$